

# CIRO

Crisis Intelligence & Response Orchestrator

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GOOGLE ANTIGRAVITY HACKATHON 2026

## Technical Architecture Document

Challenge 3 — Crisis Intelligence & Response Orchestrator (CIRO)

#AISeekho2026

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### TEAM

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# 01 PROJECT OVERVIEW

CIRO is a real-time, multi-agent AI crisis intelligence system that monitors the global and Pakistani urban environment for emerging disasters. It continuously ingests signals from social media platforms, news RSS feeds, video platforms, weather APIs, traffic APIs, and official meteorological sources. When enough corroborating signals cluster around a location, the system autonomously detects a crisis, scores its severity and impact, generates a coordinated response plan, and simulates execution of those responses — all visible in real time on an interactive world map inside a Flutter Android app.

CIRO is not a reporting dashboard. It is an autonomous decision-making system. Signals flow in, agents reason, crises appear on the map, responses execute — without human intervention. The system operates in two modes: LIVE (real signal pollers against real APIs) and SIMULATED (pre-scripted sequences through the identical agent pipeline). Both modes are available via a persistent status bar in the Flutter app. The agent pipeline is completely identical in both modes — only the signal origin differs.

CHALLENGE

Challenge 3

CIRO

PLATFORM

Google

Antigravity IDE

DEPLOYMENT

Google Cloud

asia-south1

MOBILE

Flutter 3.x

Android APK

AGENT LLM

Vertex AI

Gemini 1.5 Pro

SIGNAL SOURCES

6 Live +

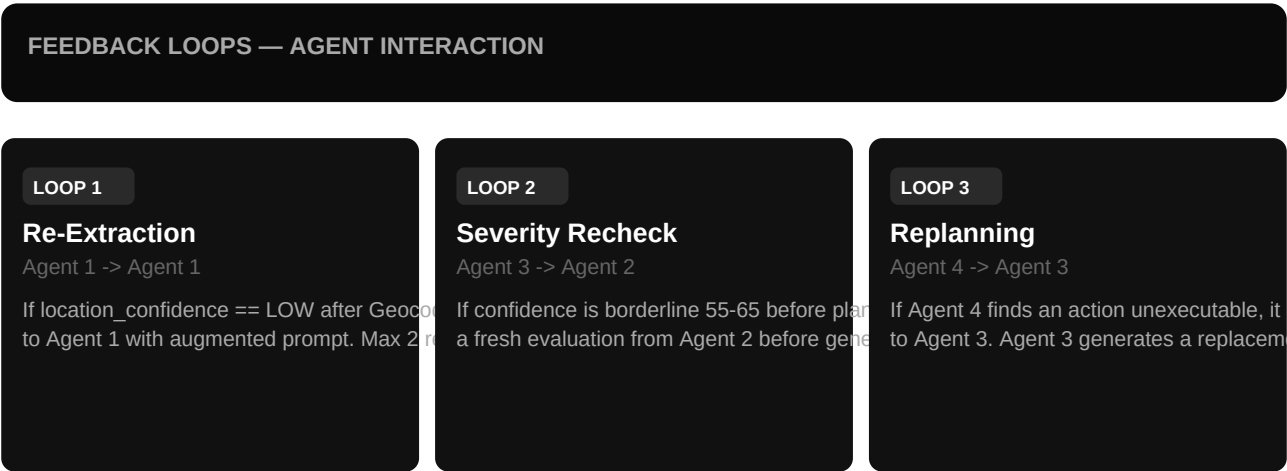
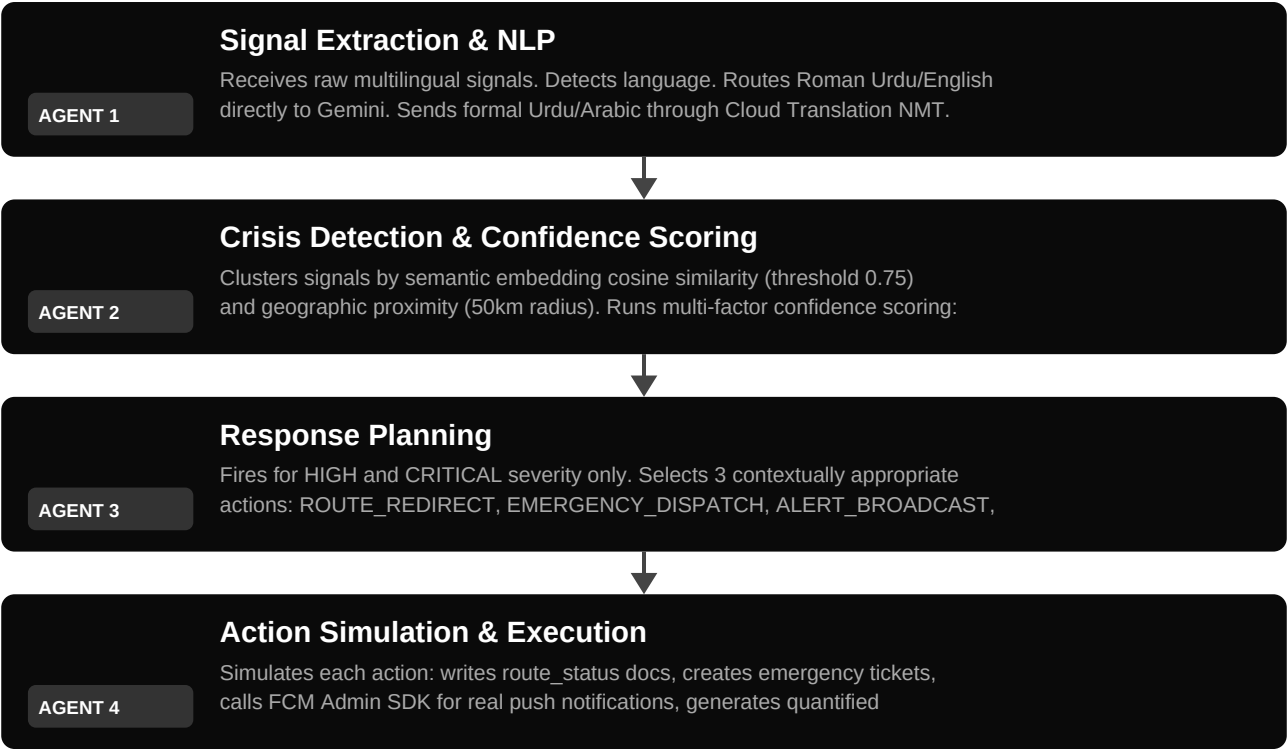
Simulated fallbacks

SYSTEM ARCHITECTURE

SIGNAL LAYER > Bluesky · YouTube · RSS · OpenWeatherMap · Routes API · PMD · Manual  
|  
LANGUAGE LAYER > Language Detection > Roman Urdu/English direct to Gemini · Formal Urdu/Arabic via NMT  
|  
INTELLIGENCE LAYER > Vertex AI Agent Builder + Cloud Workflows (Agent 1 > Agent 2 > Agent 3 > Agent 4)  
|  
DATA LAYER > Cloud Firestore (live state) · BigQuery (analytics) · Cloud Storage (artifacts)  
|  
PRESENTATION LAYER > Flutter Android App > Google Maps · Real-time pins · Response log · Agent Trace  
|  
EVENT ROUTING > Pub/Sub (6 topics) · Eventarc Advanced · Cloud Scheduler

## 02 AGENTIC PIPELINE

The agent pipeline is orchestrated by Vertex AI Agent Builder with Cloud Workflows as the step sequencer. Each agent is a Gemini 1.5 Pro call with function calling enabled, wrapped in a Cloud Workflows step. The execution graph is visible in Cloud Console and exported as a submission artifact. Three conditional feedback loops create genuine agent interaction — not a rigid waterfall.



### CLOUD WORKFLOWS ORCHESTRATION

The pipeline runs as a structured YAML workflow. Each agent step is a Cloud Workflows call step with explicit input/output mapping. Conditional switch statements implement the three feedback loops. The execution graph in Cloud Console shows SUCCESS / RUNNING / FAILED per node — screenshots of this graph are submitted as judge artifacts. Total pipeline duration from signal arrival to action execution typically runs 45–90 seconds end-to-end.

# 03 SIGNAL SOURCES

CIRO ingests from six independent signal sources, each with a real implementation and an identical-schema simulated fallback. The pipeline cannot distinguish between live and simulated signals — only the `is_simulated` flag on the Firestore document differs. Every source publishes to a dedicated Pub/Sub topic.

SOCIAL

BLUESKY

AT Protocol public API. No auth required.  
Keyword search in EN/UR/AR/ES.

Medium

MEDIA

YOUTUBE

Data API v3. Free 10k units/day.  
Video metadata + view velocity.

High

NEWS

RSS FEEDS

Dawn, Geo, ARY, BBC, Reuters, AP.  
Credibility-tiered scoring.

Very High

WEATHER

OPENWEATHERMAP

Free tier, 1000 calls/day.  
50 monitored cities globally.

High

TRAFFIC

ROUTES API

Google Maps congestion ratio.  
Monitored road corridors.

High

OFFICIAL

PMD

Pakistan Met Dept scraper.  
Highest credibility weight.

V. High

## LANGUAGE HANDLING STRATEGY

Every signal passes through language detection (Cloud Translation detect-only API, ~10ms) before any further processing. Based on detected language, one of two paths is taken:

**Roman Urdu / English** — `raw_text` sent directly to Agent 1. No translation. NMT models have minimal Roman Urdu training data and produce unreliable output. Gemini 1.5 Pro understands Pakistani internet language natively. Agent 1 prompt explicitly acknowledges the language context.

**Formal Urdu / Arabic / Other** — Cloud Translation NMT produces `normalized_text`. Both `raw_text` AND `normalized_text` are sent to Agent 1. Raw is treated as ground truth. Translation is a comprehension aid only, not the primary input. Semantic embedding is computed on `normalized_text` where available.

Firestore stores both fields always: `raw_text` (displayed in Flutter UI — judges see authentic multilingual content) and `normalized_text` (used internally by agents). Every signal is embedded via Vertex AI Text Embeddings API (text-embedding-004) for semantic clustering in Agent 2.

# 04 CONFIDENCE SCORING ENGINE

The confidence scoring engine runs inside Agent 2. It determines whether a signal cluster constitutes a real crisis and how severe that crisis is. Signals are grouped by semantic embedding cosine similarity (threshold 0.75) and geographic proximity (50km radius) — not by keyword matching. This means "boat chahiye Lahore mein" clusters with flood signals because its embedding is semantically close, even though it shares no disaster keywords.

| BASE SIGNAL SCORES               |     | CLUSTERING BONUSES        |     |
|----------------------------------|-----|---------------------------|-----|
| PMD Official Advisory            | +40 | 3+ signals same city      | +12 |
| Major Pakistani News (Dawn/Geo)  | +28 | 3+ signals same district  | +22 |
| International News (BBC/Reuters) | +25 | 5+ signals same district  | +35 |
| YouTube 100k+ views              | +35 | 8+ signals same district  | +50 |
| YouTube any video                | +15 | All signals within 15 min | +25 |
| OpenWeatherMap Alert             | +30 | Frequency increasing      | +12 |
| Traffic spike >4x baseline       | +28 | 4+ different source types | +28 |
| Bluesky social post              | +10 | YouTube video exists      | +15 |
| Manual GPS report                | +18 | Video >10k views          | +10 |



**CONFIDENCE DECAY**

A `decay_scheduler` Cloud Function runs every 60 minutes via Cloud Scheduler. It queries all crisis documents with status `!= RESOLVED` and `updated_at < now - 1hr`. For each: subtract 5 confidence points. Below 30 -> `RESOLVING`. Below 15 -> `RESOLVED`. This prevents stale crises from persisting indefinitely. A crisis with no new corroborating signals closes itself within 3-6 hours.

# 05 GOOGLE CLOUD TECHNOLOGY STACK

## DEVELOPMENT

### Antigravity IDE

Primary dev environment.  
All code written here.  
Generates judge artifacts.

## ORCHESTRATION

### Vertex AI Agent Builder

Formal multi-agent workflow.  
Traceable execution logs.  
Judge submission evidence.

## ORCHESTRATION

### Cloud Workflows

Step sequencer with YAML.  
Conditional feedback loops.  
Visible execution graph.

## LLM

### Gemini 1.5 Pro

Powers all 4 agents.  
Function calling for typed output.  
Chain-of-thought prompting.

## SEMANTICS

### Text Embeddings API

text-embedding-004.  
Semantic signal clustering.  
Cosine similarity in Agent 2.

## LANGUAGE

### Cloud Translation

Detect-only for Roman Urdu.  
NMT for formal Urdu/Arabic.  
Normalized\_text as hint only.

## BACKEND

### Cloud Run

Python 3.12 container.  
Hosts 4-agent pipeline.  
IAM-authenticated.

## POLLERS

### Cloud Functions

10 functions total.  
One per signal source.  
Plus decay + simulation.

## MESSAGING

### Pub/Sub

6 dedicated topics.  
One per signal source.  
Eventarc routes messages.

## ROUTING

### Eventarc Advanced

Pre-pipeline filtering.  
Direct Cloud Workflows trigger.  
Named pipeline resources.

## TRIGGERS

### Cloud Scheduler

Fires all pollers.  
Per-source cadence 5-30 min.  
Decay scheduler every 60 min.

## DATABASE

### Cloud Firestore

Live crisis state.  
Real-time Flutter stream.  
Signals subcollection.

## ANALYTICS

### BigQuery

Historical crisis data.  
Firestore streaming export.  
Gemini NL SQL queries.

## ARTIFACTS

### Cloud Storage

Antigravity artifact bucket.  
Simulated feed JSONs.  
Demo scenario sequences.

## MAPS

### Google Maps SDK

Flutter Android integration.  
Custom severity markers.  
Traffic layer overlay.

## MAPS

### Directions API

Route redirect polylines.  
Before/after visualization.  
Alternate route rendering.

## TRAFFIC

### Routes API

Congestion ratio per corridor.  
Monitors 50+ road segments.  
Simulated fallback schema.

## MAPS

### Geocoding API

Informal location -> lat/lng.  
Pakistani neighbourhood names.  
City-level fallback.

## ALERTS

### Firebase FCM

Real push notifications.  
Agent 4 Admin SDK call.  
Deep links to crisis route.

## SECURITY

### Secret Manager

All API keys stored here.  
Injected at Cloud Run runtime.  
No keys in source code.

## OBSERVABILITY

### Cloud Logging

Structured JSON agent logs.  
Exported as judge artifacts.  
Cloud Trace call graph.

## CI/CD

### Artifact Registry

Docker image storage.  
Cloud Build auto-deploy.  
Push-to-main trigger.

## 06 ANTIGRAVITY — DEVELOPMENT & ARTIFACTS

### WHAT ANTIGRAVITY DOES IN CIRO

Antigravity is the primary development environment — not supplementary. All CIRO code is written, planned, and deployed through Antigravity's Agent Manager. Its Gemini 3.1 Pro backbone provides a 2-million-token context window, enabling the entire codebase (Flutter + Python backend + IaC scripts) to be reasoned about holistically.

#### Key capabilities used in CIRO:

Cloud Run MCP server — deploys backend directly from IDE without manual gcloud commands. Parallel agent execution — up to 5 agents working simultaneously across Flutter, backend, IaC, and test files. Plan artifacts generated before every execution — saved automatically to `_antigravity_artifacts/`. Walkthrough artifacts capture what was built, commands run, and verification results after each mission.

### ARTIFACT STRATEGY FOR JUDGES

The challenge requires Agent Trace / Logs covering Workplan, Tasks Plan, reasoning steps, decision flow, and action execution. CIRO satisfies this with four artifact sources:

#### `_antigravity_artifacts/plans/`

Implementation plan artifacts — one per mission, auto-generated.

#### `_antigravity_artifacts/traces/`

Agent reasoning trace exports — one per major feature.

#### `_antigravity_artifacts/walkthroughs/`

Build summaries — what was built, commands run, verifications.

### Cloud Workflows Execution Graphs

Visual proof of multi-agent orchestration — screenshots submitted.

### 27 ANTIGRAVITY MISSIONS — BUILD SPRINT

01 Project Scaffold · 02 Pub/Sub + Eventarc · 03 Bluesky Poller · 04 YouTube Poller · 05 RSS Poller · 06 Weather + Traffic + PMD Pollers · 07 Language Detection · 08 Cloud Translation · 09 Text Embeddings · 10 Cloud Workflows Orchestrator · 11 Agent 1 Extraction · 12 Agent 2 Detection + Scoring + Impact · 13 Agent 3 Response Planning · 14 Agent 4 Action Simulation · 15 Decay Scheduler · 16 Gemini Search Grounding · 17 Flutter Map Screen · 18 Flutter Crisis Detail Sheet · 19 Flutter Signal Feed · 20 Flutter Report Disaster · 21 Flutter Analytics + BigQuery · 22 Flutter Agent Trace Screen · 23 Flutter Simulation Mode · 24 Demo Scenarios Seed · 25 FCM Push Notifications · 26 Demo Video Prep · 27 README + Submission

## 07 DATA SCHEMA & SIMULATION MODE

### FIRESTORE CRISIS EVENT SCHEMA

```
crisis_id · type · city · district ·  
country lat · lng · severity · confidence ·  
status signal_count · signal_source_summary  
· impact_analysis impact_metrics ·  
agent_reasoning · actions_planned  
actions_executed · route_changes ·  
dispatch_tickets broadcast_logs ·  
system_state_before · system_state_after  
created_at · updated_at · resolved_at
```

Signals stored in subcollection

crisis\_events/{id}/signals — not embedded  
array. Prevents 1MB document limit. Parent  
holds signal\_count and  
signal\_source\_summary for fast display.  
Flutter lazy-loads subcollection only on  
Contributing Signals expand.

### SIMULATION MODE

Simulation mode is a first-class feature. A persistent status bar on the Map screen shows LIVE SIGNALS or SIMULATED as a pill button. The Flutter app has a scenario selector bottom sheet with 5 pre-scripted demo scenarios.

#### How it works:

A simulation\_replay Cloud Function reads the selected scenario JSON from Cloud Storage. It replays signals one by one into the correct Pub/Sub topics with realistic time delays. The full agent pipeline runs identically — Agent 1 through Agent 4 — exactly as in live mode. Simulated signals carry is\_simulated: true in Firestore and display a SIM badge in the app. By signal 3 (~90 seconds), confidence typically crosses 45 and the map pin drops live.

The pipeline has zero awareness of simulation mode. This means the demo is a genuine demonstration of the real system — not a mocked UI.

## 08 EVALUATION CRITERIA MAPPING

### 25%

#### USE OF GOOGLE ANTIGRAVITY

Antigravity is the sole development environment. All 27 missions executed inside it. Cloud Run deployed via Cloud Run MCP server. Every mission generates plan, walkthrough, and trace artifacts stored in `_antigravity_artifacts/` and submitted directly to judges. Vertex AI Agent Builder (accessed via GCP integration) hosts the formal agent workflow.

### 20%

#### AGENTIC REASONING & COORDINATION

Four agents with single responsibilities and explicit typed inputs/outputs. Cloud Workflows provides a visible orchestration graph. Three conditional feedback loops create genuine agent interaction. Agents communicate via Firestore shared state. All reasoning logged in structured JSON in `agent_reasoning` array on every crisis document.

### 20%

#### SITUATION DETECTION & ANALYSIS

Seven independent signal sources with different credibility weights. Multi-factor confidence scoring with geographic, temporal, source diversity, media, and traffic bonuses. Semantic embedding clustering by meaning not keywords. Multilingual handling: Roman Urdu/English direct to Gemini, formal Urdu/Arabic via NMT. Confidence decay handles resolution.

### 15%

#### ACTION PLANNING & SIMULATION

Five action types: `ROUTE_REDIRECT`, `EMERGENCY_DISPATCH`, `ALERT_BROADCAST`, `RESOURCE_ALLOCATION`, `HOSPITAL_PREPAREDNESS`. Agent 3 selects contextually. Agent 4 produces real Firestore state changes, real FCM notifications, real Directions API polylines. Quantified before/after `impact_metrics` with specific numbers. Replanning loop if action unexecutable.

# 10%

## TECHNICAL IMPLEMENTATION

Clean layered architecture. Eventarc Advanced for pre-pipeline filtering. Cloud Workflows for traceable orchestration with conditional branches. BigQuery for historical analytics. Signals subcollection prevents Firestore 1MB limit. Secret Manager for all credentials. Confidence decay via scheduled Cloud Function. Simulated fallbacks for all external sources.

# 10%

## INNOVATION & UX

Global + Pakistan dual-view map. Raw multilingual signals with language badges in the detail sheet. Interactive Agent Trace screen inside the app. Simulation mode with signal health dashboard. Semantic embedding clustering. Traffic layer integrated into map UI. Natural language BigQuery queries in the analytics screen.

## 09 DEMO SCENARIOS & FLUTTER APP

### SCENARIO 1

#### Lahore Flash Flood

CRITICAL 94%

Gulberg, Lahore, PK

3 Bluesky Roman Urdu posts + Geo News RSS + OWM 72mm/hr rainfall  
+ PMD Punjab advisory + Canal Road 4.2x congestion

Actions: Route redirect Canal->Jail Rd, 3 rescue units, FCM alert 50k users

### SCENARIO 2

#### Karachi Road Blockage

HIGH 71%

Saddar, Karachi, PK

2 Bluesky posts + ARY News RSS + YouTube video 18k views

Actions: Alternate route M.A. Jinnah Rd, traffic warden dispatch

### SCENARIO 3

#### Islamabad Heatwave

MEDIUM 58%

G-10, Islamabad, PK

2 English social posts + OWM temperature spike 46C

Actions: Advisory only — FCM public advisory, hospital preparedness

### SCENARIO 4

#### California Wildfire

CRITICAL 88%

Los Angeles County, CA, USA — GLOBAL DEMO

5 Bluesky posts + 2 Reuters/AP RSS + YouTube 120k views

Actions: Evacuation route redirect, emergency dispatch, alert broadcast

**SCENARIO 5****Istanbul Earthquake**

HIGH 79%

Istanbul, Turkey — GLOBAL DEMO

3 Turkish social posts (NMT translated) + BBC RSS

Actions: Hospital preparedness, emergency dispatch, resource allocation

**10 FLUTTER APP — 12 SCREENS****S1 ONBOARDING**

First launch only. 4-slide carousel explaining CIRO, signal-to-response flow, Pakistan focus, and simulation mode. Skippable.

**S2 MAP**

Hero screen. Global (hybrid satellite) and Pakistan (normal + traffic layer) views. Custom severity-coded markers. Status bar with LIVE/SIM toggle and source health dots.

**S3 CRISIS DETAIL SHEET**

DraggableScrollableSheet on pin tap. 35%-85% height. Shows impact summary, confidence bar, signal/action pills. Zoom to City button.

**S4 CRISIS DEEP DIVE**

Full screen. 6 sections: Situation, Impact Analysis, Before/After Metrics, Contributing Signals (raw multilingual text with language badges), Response Actions, Agent Reasoning Timeline.

**S5 SIGNAL FEED**

Reverse-chronological incoming\_signals stream. Real-time Firestore listener. Filter chips by source type. Shows raw text in original language.

**S6 AGENT TRACE**

Interactive 4-node execution graph. Tap any node to expand input/output/reasoning. Show Raw JSON toggle. Crisis selector dropdown. Execution timeline bar.

**S7 ANALYTICS**

BigQuery-powered. Natural language query bar with example chips. Dashboard cards: totals, by type, top cities, signal sources, response times.

**S8 REPORT DISASTER**

Manual signal submission. Any language text input. GPS or pick-on-map location. Optional photo. Direct POST to Agent 1 bypassing Pub/Sub.

**S9 NOTIFICATIONS**

FCM alert history. Unread indicators. Full alert message in English and Urdu. Deep link to crisis on card tap.

**S10 SETTINGS**

Mode toggle, simulation scenario selector, source health rows with last poll timestamps, Force Refresh All, About section with artifact bucket link.

**S11 SIM BOTTOM SHEET**

Opens from status bar tap. Scenario radio list with detection time estimates. Start/Stop simulation. Switch back to Live link.

**S12 MORE MENU**

Secondary nav hub. Links to Report, Disaster Directory, Analytics, Notifications, Settings, About.